Power Analyses Introduction

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PART 1: Introduction

PART 2: Classic Statistic Power Analyses

PART 3: Mont-Carlo Simulation

PART 1: Introduction

What is statistical power?

Statistical power is the probability that your statistical test correctly detects a real effect in the population.

"If an effect is real out there in the population, how likely am I to catch it with my sample and my test?"

Sample size

Model

PART 1: Introduction

Why we want to do power analyses?

1. After collecting data, determine whether the *design/analysis* and *sample size* had sufficient power to detect relationships if they were to exist.

2. Before collecting data, determine the *necessary sample size* to achieve sufficient power (e.g., power $\geq$ .80).

3. Before collecting data, determine the *minimally detectable effect size (MDES)* for a range of sample sizes.

PART 1: Introduction

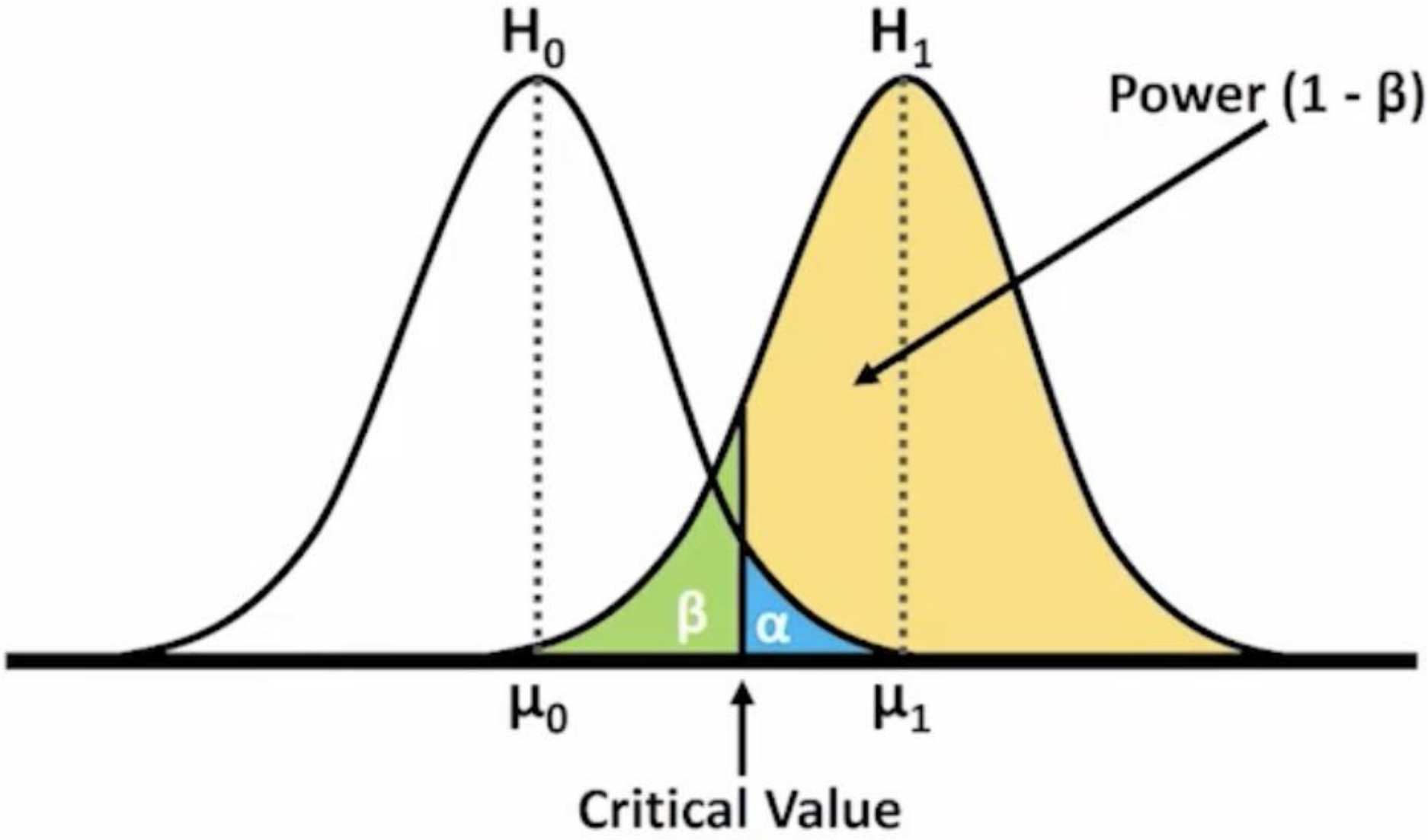
	H_0 True	H_A true
Reject H_0	Type I error	Correct decision
Fail to reject H_0	Correct decision	Type II error

Type I error occurs when you reject a hypothesis that is true. Probability of making a Type I error is α (the significance level). $\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true})$

Type II error occurs when you fail to reject a false null hypothesis. Probability is making a Type II error is β. $\beta = P(\text{Fail to Reject } H_0 \mid H_a \text{ true})$

Power of the test is the probability of rejecting H_0 when H_A is true. i.e. Power is the probability of finding an effect if it exists. $\text{Power} = 1 - \beta = P(\text{Reject } H_0 \mid H_a \text{ true})$

PART 1: Introduction



PART 1: Introduction

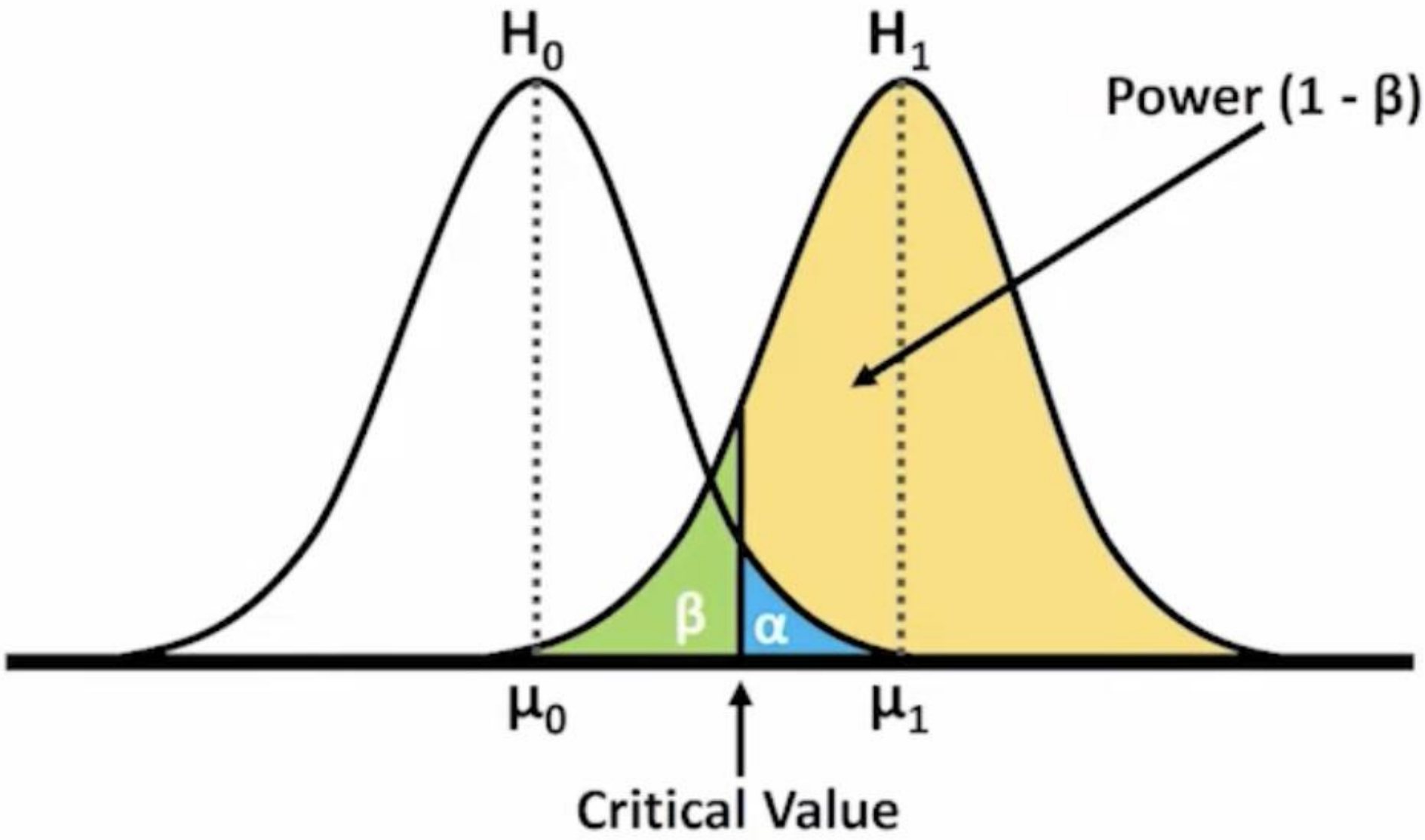
Technique	Core idea
Analytic	Power derived from known sampling distributions
Monte Carlo simulation	Simulate many full studies and count detections

PART 2: Classic Statistic Power Analyses

"Power of the test is the probability of rejecting H_0 when H_A is true. (Power is the probability of finding an effect if it exists)."

"If we repeatedly sampled from the population, the test statistic (mean, sd, relationship) would follow a known probability distribution. Then we can compute the probability of rejection directly from that distribution."

PART 2: Classic Statistic Power Analyses



PART 2: Classic Statistic Power Analyses

Cola Example

Imagine we are a Cola factory….

Let X be the volume of cola in a bottle (in milliliters). The population standard deviation is known to be 3 ml. We take a simple random sample of size $n = 25$ and compute the sample mean $\overline{X}$.

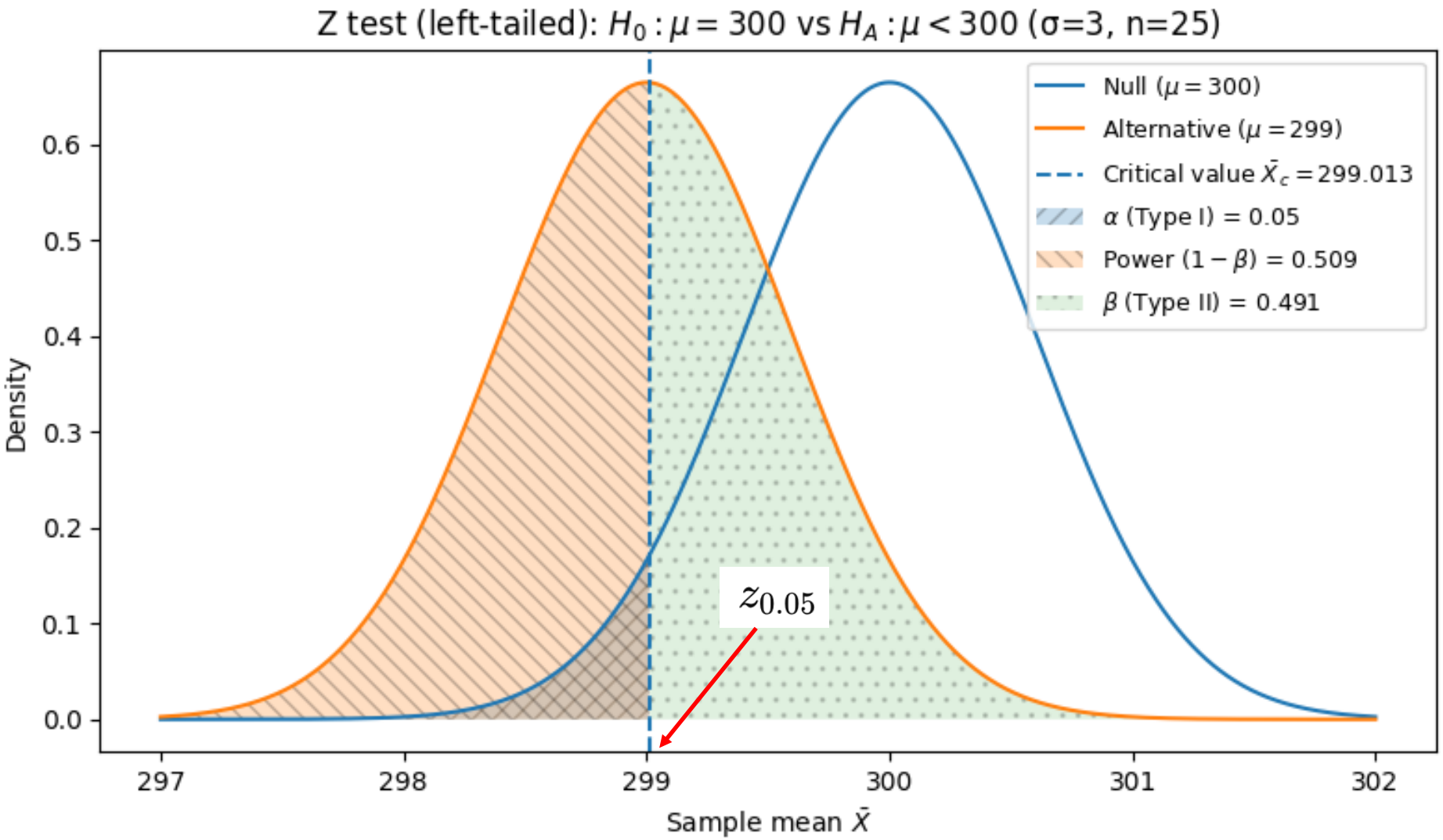
We conduct a **one-sample, left-tailed z-test** at the $\alpha = 0.05$ level to test:

$$H_0: \mu = 300 \; vs. H_A: \mu < 300$$

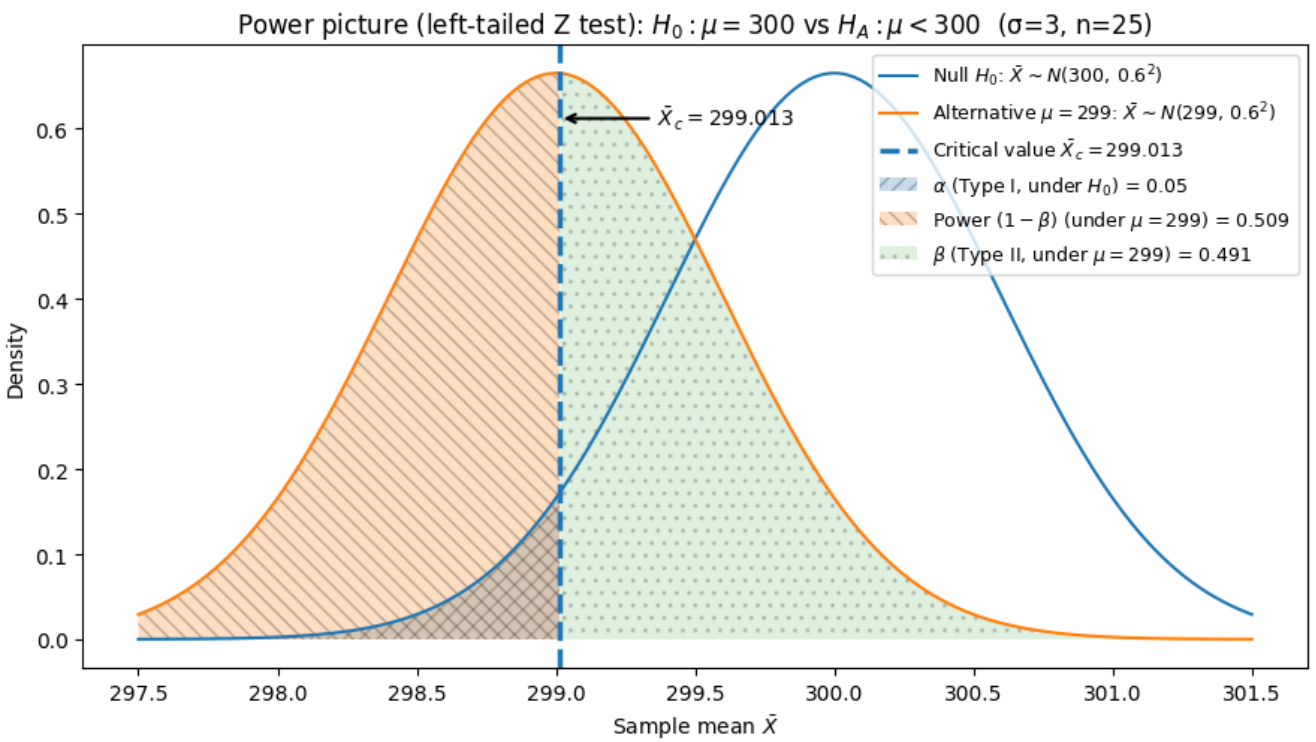
Question:

If the true population mean is actually $\mu = 299$, what is the **power of this test** (that is, the probability that this z-test will correctly reject H_0?)

PART 2: Classic Statistic Power Analyses



PART 2: Classic Statistic Power Analyses

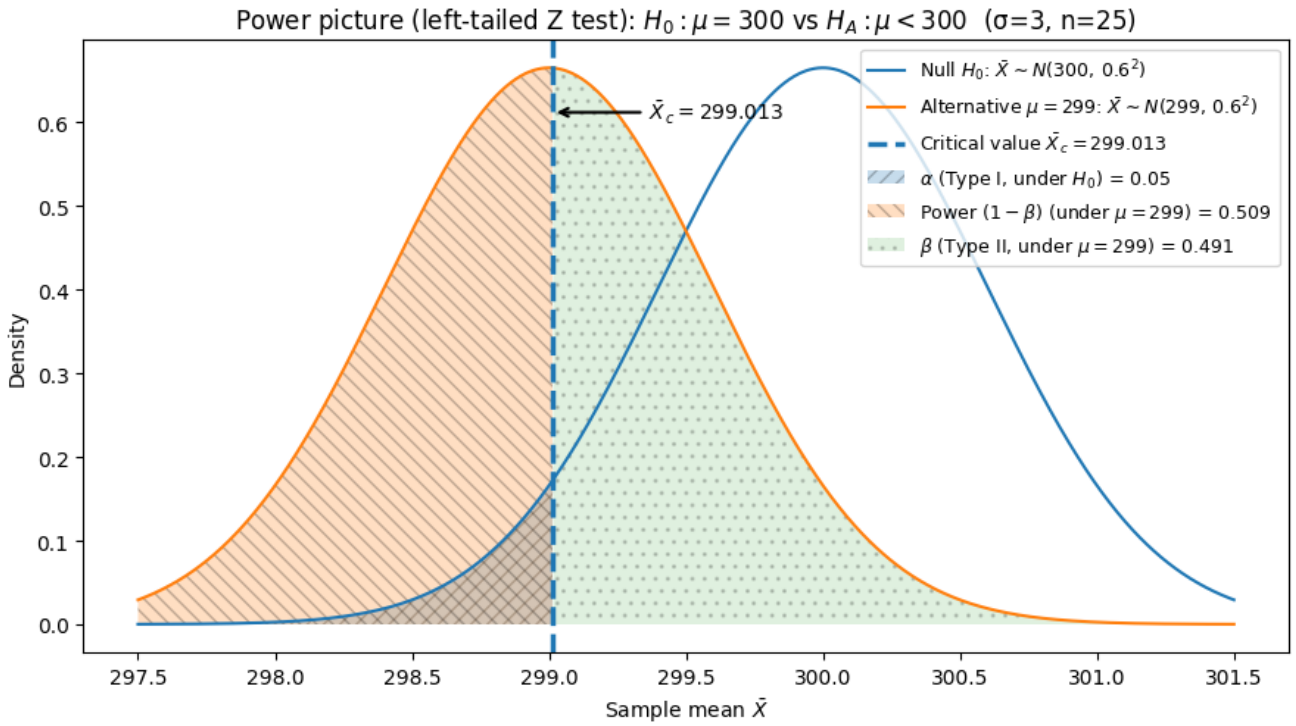


$$\bar{X} \sim N\left(\mu,\ \frac{\sigma^2}{n}\right), \qquad \sigma = 3,\ n = 25 \Rightarrow \mathrm{SE} = \frac{\sigma}{\sqrt{n}} = \frac{3}{5} = 0.6$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\bar{X} - 300}{0.6} \qquad z_{0.05} = -1.645$$

$$\frac{\bar{X} - 300}{0.6} < -1.645 \quad \Rightarrow \quad \bar{X} < 300 - 1.645(0.6) = 300 - 0.987 = 299.013$$

PART 2: Classic Statistic Power Analyses



Power $= P(\text{reject } H_0 \mid \mu = 299)$:

$$\text{Power} = P(\bar{X} < 299.013 \mid \mu = 299)$$

Under $\mu = 299$, $\bar{X} \sim N(299, 0.6^2)$. Standardize:

$$\text{Power} = P\left(\frac{\bar{X} - 299}{0.6} < \frac{299.013 - 299}{0.6}\right) = \Phi\left(\frac{0.013}{0.6}\right) = \Phi(0.0218) \approx 0.5087$$

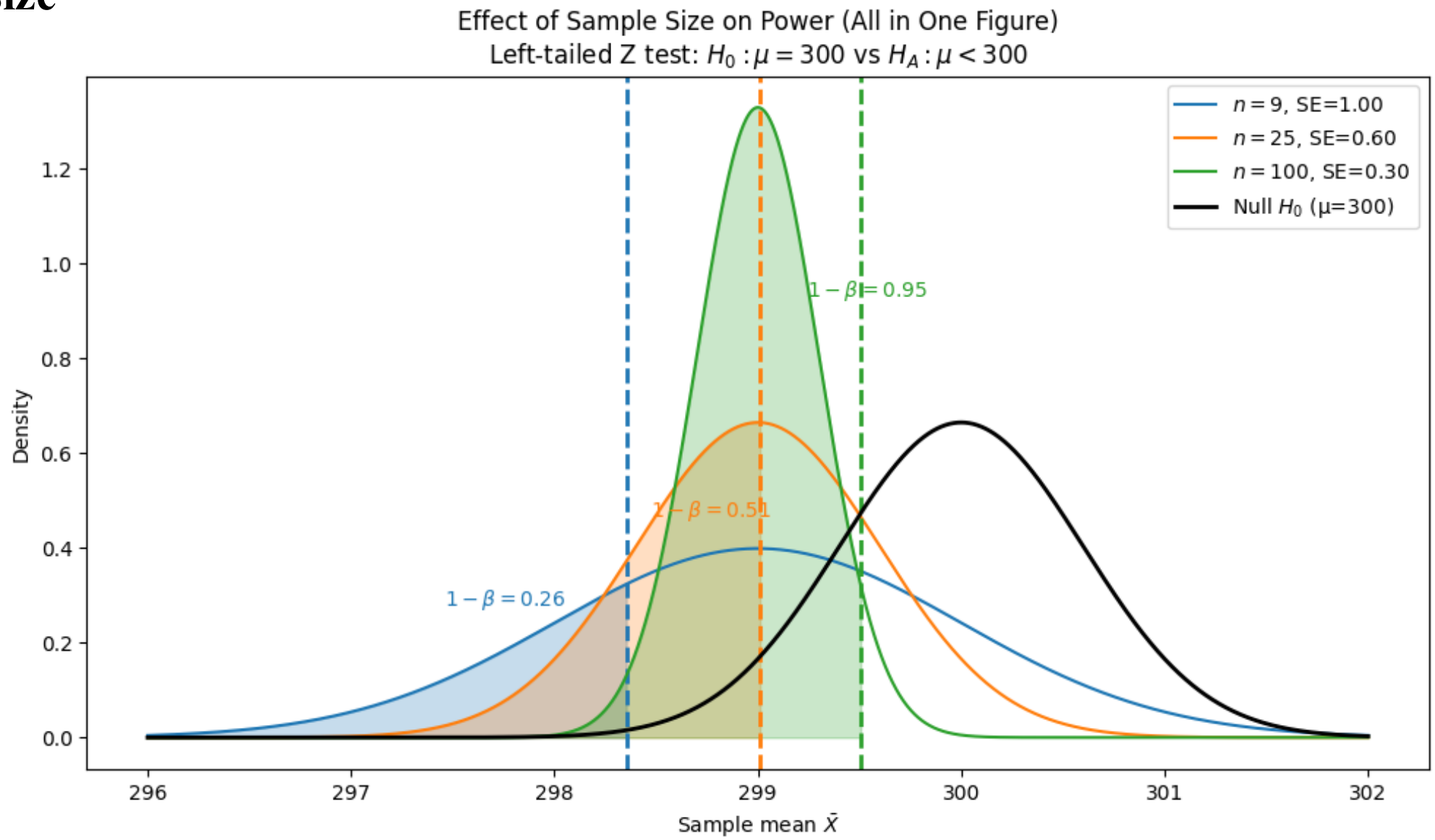
$$\boxed{\text{Power at } \mu = 299 \ \approx \ 0.509}$$

Three important elements

1. **Sample size**
2. **Distribution & Alpha**
3. **Effect size (e.g., r, β, cohen's d)**

PART 2: Classic Statistic Power Analyses

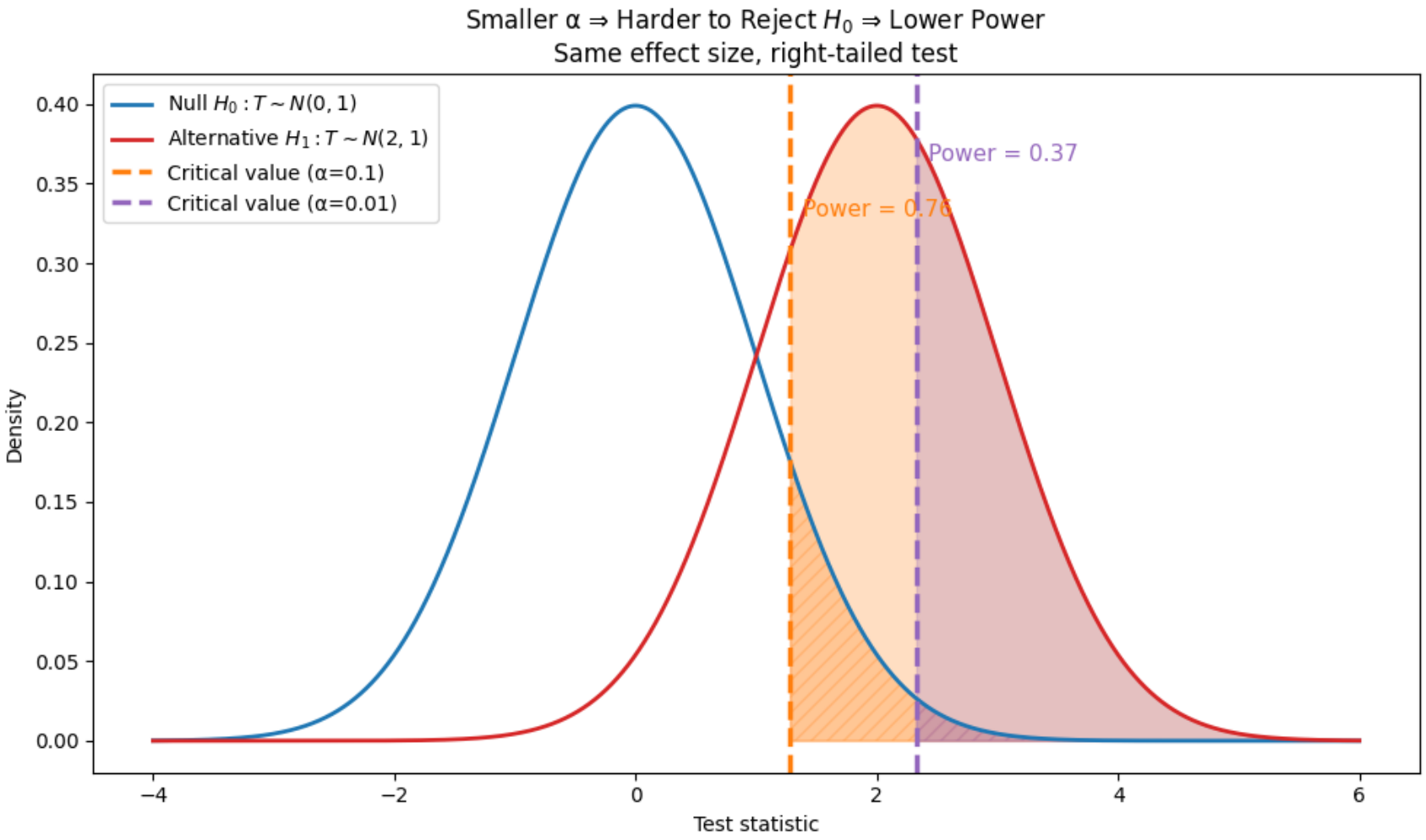
1. Sample size



PART 2: Classic Statistic Power Analyses

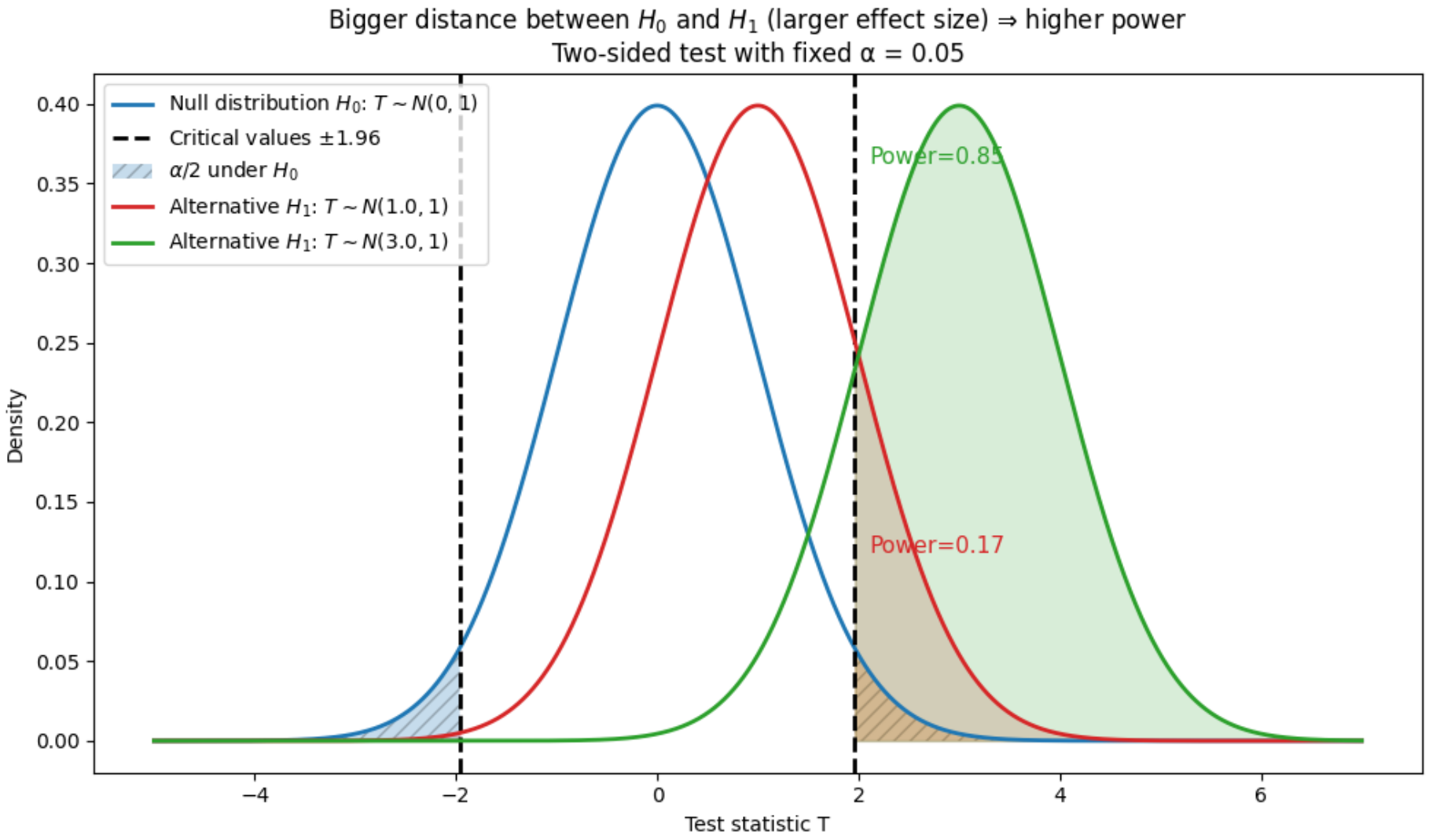
2. Distribution & α value

The smaller α, the less likely a test is to reject H_0, and thus the less power our tests have.



PART 2: Classic Statistic Power Analyses

3. “Distance”: true effect size



PART 2: Classic Statistic Power Analyses

Advantages

Computational cost is low

Limitation

Highly depends on the assumption of distribution

Model complexity

Cannot handle missing data

Mont Carlo simulation

PART 3: Mont-Carlo Simulation

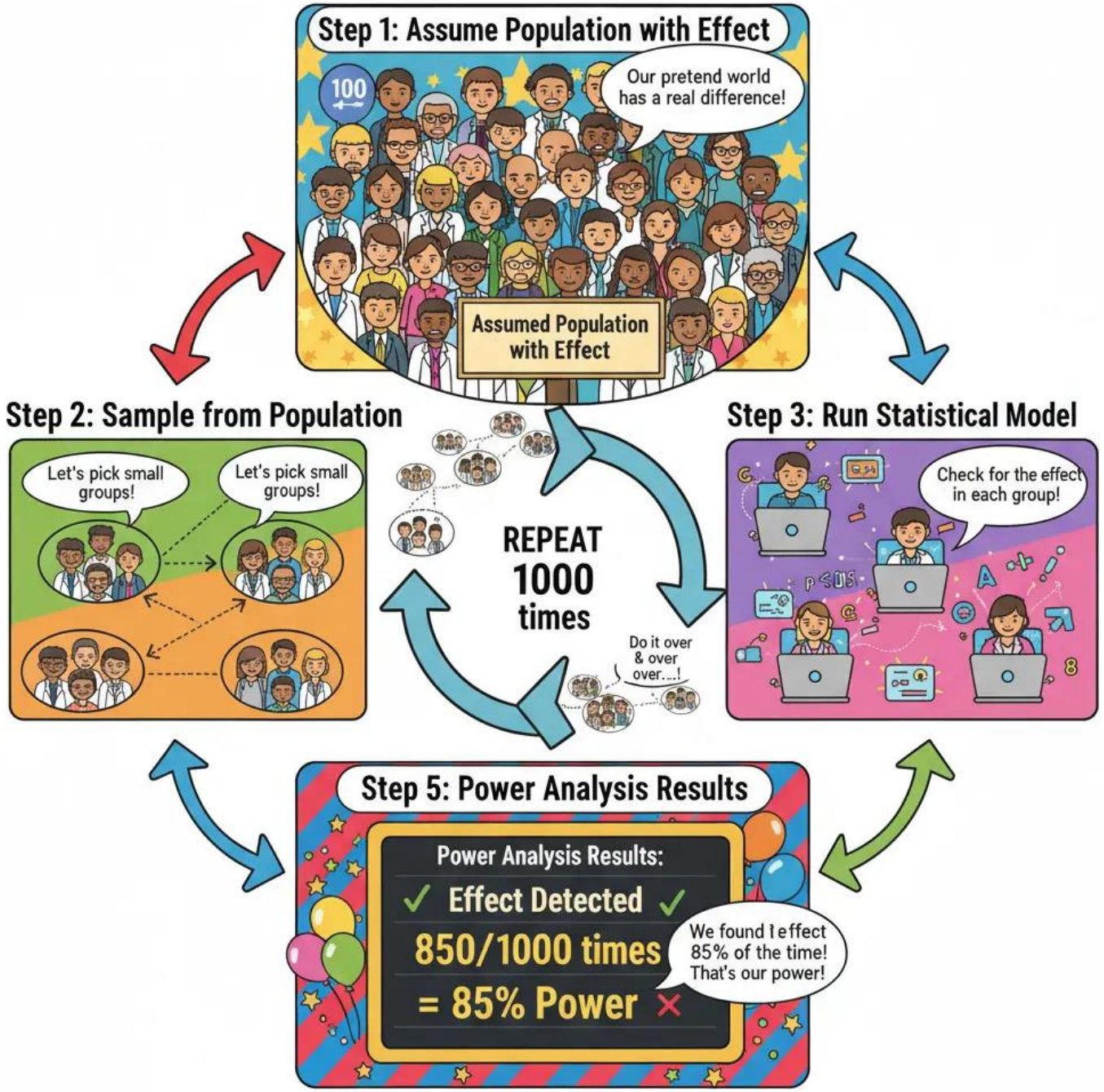
"Power of the test is the probability of rejecting H_0 when H_A is true. (Power is the probability of finding an effect if it exists)."

~~"If we repeatedly sampled from the population, the test statistic would follow a known probability distribution. Then we can compute the probability of rejection directly from that distribution."~~

Let's actually do the sampling process and count how many times we can find the effect !

Muthén, L. K., & Muthén, B. O. (2002). How to use a Monte Carlo study to decide on sample size and determine power. *Structural equation modeling*, *9*(4), 599-620.

PART 3: Mont-C


Step 1: Assume Population with Effect
100
Our pretend world has a real difference!
Assumed Population with Effect
Step 2: Sample from Population
Let's pick small groups!
Let's pick small groups!
REPEAT 1000 times
Do it over & over over....!
Step 3: Run Statistical Model
Check for the effect in each group!
Step 5: Power Analysis Results
Power Analysis Results:
✓ Effect Detected ✓
850/1000 times
= 85% Power ✕
We found 1 effect 85% of the time! That's our power!

PART 3: Mont-Carlo Simulation

Simsem

https://simsem.org/

pwrSEM

Wang, Y. A., & Rhemtulla, M. (2021). Power analysis for parameter estimation in structural equation modeling: A discussion and tutorial. *Advances in Methods and Practices in Psychological Science*, *4*(1), 2515245920918253.

https://github.com/yilinandrewang/pwrSEM

Thanks!

When You Do a Proper Power Analysis
Significant Results!
- Publish Paper
- Celebrate!
SCIENCE SUCCESS!
When You Don't Do a Power Analysis
No Significant Results
- Can't Publish
- Wasted Time & Money...
SADNESS & REGRET